

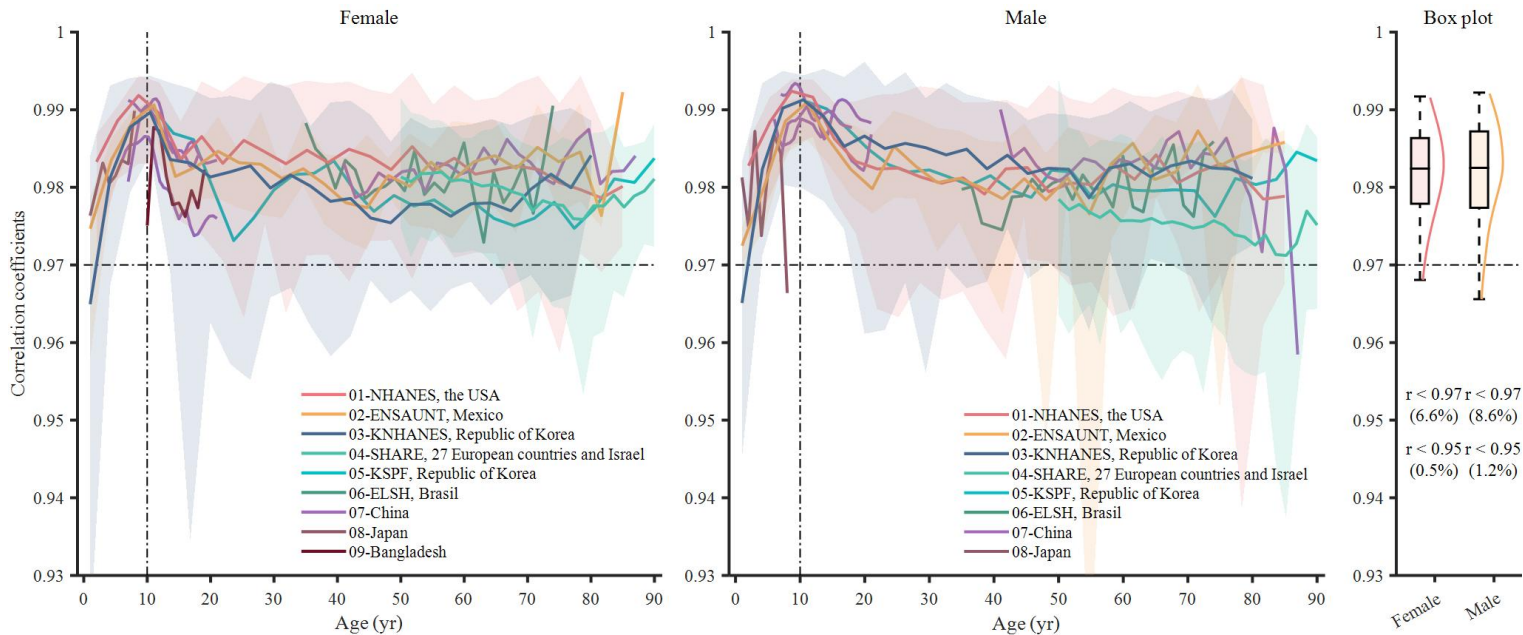


Fig. 1. Life-course trajectories of linear correlation coefficients between weight and weight×height. For NHANES, ENSAUNT, and KNHANES, the solid line represents the results of data pooled from all survey waves; the shading shows the 5th to 95th percentile range across individual waves. For SHARE, the solid line shows the mean across all countries, with the shading indicating the 5th to 95th percentile range across individual countries. For KSPF, ELSH, China, Japan, and Bangladesh, curves were smoothed to clarify age-related trajectories. Box plots are used to display the distribution of correlation coefficients. The proportions (%) of subgroups with values below 0.97 and 0.95 are indicated below the box plots. The strong linear correlation is a key observation motivating the derivation of the hyperbolic constraint. Randomization test confirms that the correlation arises from biological coordination, thereby establishing a true constraint rather than a statistical artifact.